



A multi-player game created by **amdon** for



Ministry of Education
SINGAPORE

User Guide

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BACKGROUND

The Story so Far

DIGESTICO CORPORATION REPORT

Date

20 February 2048

Subject

GEORGE KILLINEY, Male, 18

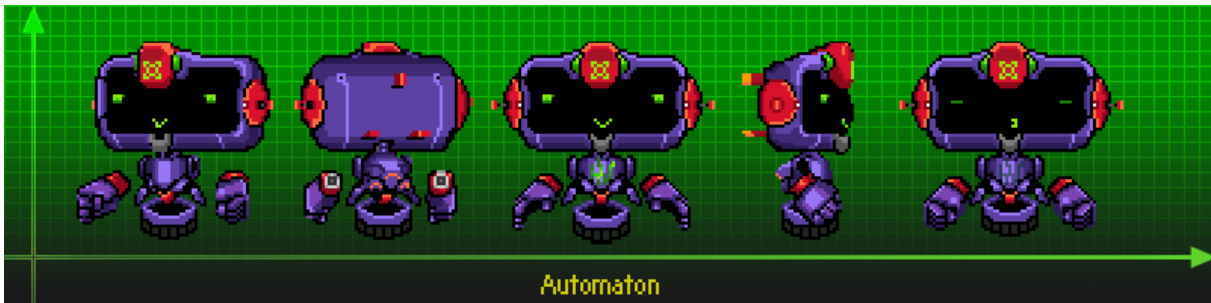
System Failure

Digestive



George Killiney had injured his head in a traffic accident. The accident left him with a serious digestive disorder. To help George, doctors have put Digestico's GUT REPLACEMENT MODULE into his body.

The GUT REPLACEMENT MODULE is made up of several control stations. It is run by five tiny, intelligent robots called Automaton.



As an Automaton, YOU have been entrusted to

- Help George digest properly
- Make sure George keeps to a healthy diet
- Solve crises within the gut.
- Keep George alive until he recovers fully

ALL YOU NEED TO KNOW ABOUT THE GUT REPLACEMENT MODULE & YOU

Off to a new Start

Welcome to your new assignment, Automaton! This guide will teach you everything you need to know about the DIGESTICO GUT REPLACEMENT MODULE. The GUT REPLACEMENT MODULE works just like George's real digestion system, with control stations that look after the different stages of digestion. We need your help to run the stations.

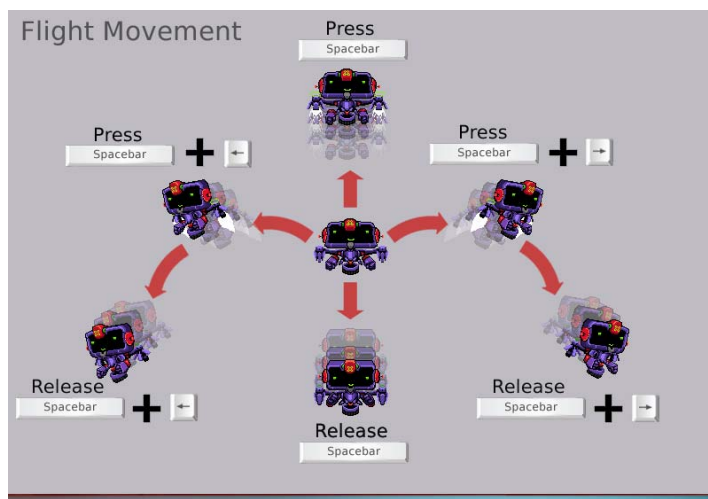
Moving Around

All Automatons are able to float or fly inside the MODULE. Use the keys you find on the keyboard to move:

How to float on the ground



How to Fly



Some things you might need to know first

Carbohydrates

Organic nutrient and complex sugar that is commonly used by our body to provide energy for our day to day activities. These need to be digested into simple sugar known as glucose for use.

Proteins

Organic nutrient used in the building and repairing of tissue and muscle.

Fats

Organic nutrient is made up of glycerol and fatty acids. Like protein, they are also essential to cell functions. Fats are harder to use as energy sources, but are better used for energy storage.

Water

Essential inorganic substance for life and responsible for many essential processes in our body. The human body is made up of 60 to 70 percent of water. We need to drink water to replenish loss through perspiration, respiration and passing out of waste.

Fibre

Substance indigestible by the human body and is taken essentially to help absorb water and ease the removal of waste products such as faeces. Cellulose, found in green vegetables that we eat is an example of dietary fibre.

Enzyme

Special type of protein that our body produces to break down food into simplest form. This process of breaking down food is known as **digestion**.

Vitamins

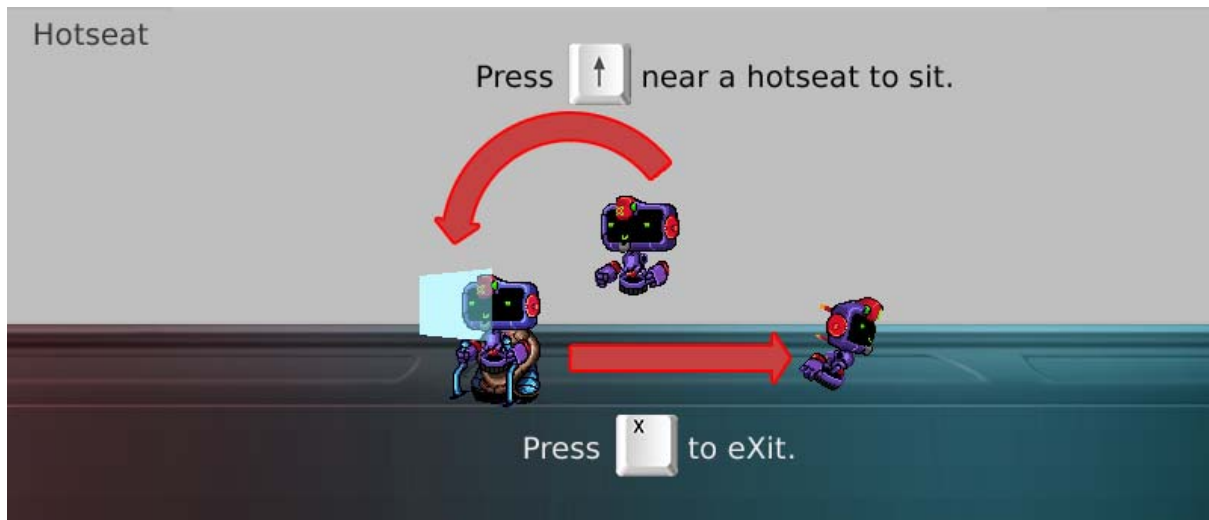
Vitamins, minerals are also essential for our body functions but they are required in small traces and therefore not covered in our game.

Hotseats



Automatons can perform many digestive activities, by sitting on hotseats found in the control stations. You may find more than 1 hotseat in a control station. The countdown timer beside the hotseat starts when you sit on the hotseat. Hotseats get a little too hot if used for 120 seconds or more. When that happens, it is best to get off and wait for the timer to go back to 120 seconds. Better yet, you are encouraged to venture to other hotseats where help is needed.

How to use hotseats



Things you will find on the screen

Information on the screen will help you in your task. Here's what your screen may look like:



We have placed numbered circles next to some useful displays. Read on to find out what these displays do.

1



Event Timer

This meter displays George's current activity. It can show the time of day, month or meal. A countdown timer below it shows how much time, in seconds, is left till the next activity.

2

George Mini-map and Navigation Aids

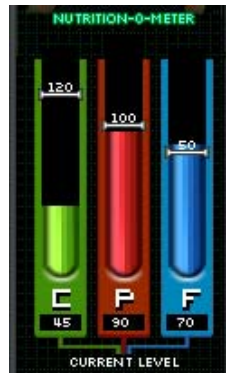
The minimap allow you to see where in George's body your team is. You are identified by player numbers assigned to you at the start of the game.

The mini-map shows 5 zones:

Zone 1:	Brain quarters, Mouth room, Oesophagus observatory, portal to Stomach.
Zone 2:	Stomach room
Zone 3:	Small intestine (turrets 1,2 & absorption room)
Zone 4 (right of zone 3):	Enzyme depot
Zone 5 (below zone 3):	Colon chamber



3



Nutrition-o-meter

The Nutrition-o-meter is very important.

The numbers at the top of the meter shows how much nutrients you need to absorb to keep George healthy. You can also see them as white coloured lines.

The numbers below indicates the amount of nutrients, carbohydrate(c), protein(p), and fat(f) absorbed so far. The coloured bars will rise when nutrients are absorbed.

Watch these closely as they determine how healthy George would become if he keeps eating like this. All numbers we see are in units called **blocks**. We will explain more on this later.

4

BLOCKS WASTED : 67

Wastage meter

The Automatons should properly digest and absorb food at all times. Otherwise, the wastage meter will increase. The wastage meter shows the number of blocks of nutrients not absorbed.

5

499958

Score & Team Contribution

The score meter shows well your team is doing. Your team's actions may cause the score to rise or fall.

6

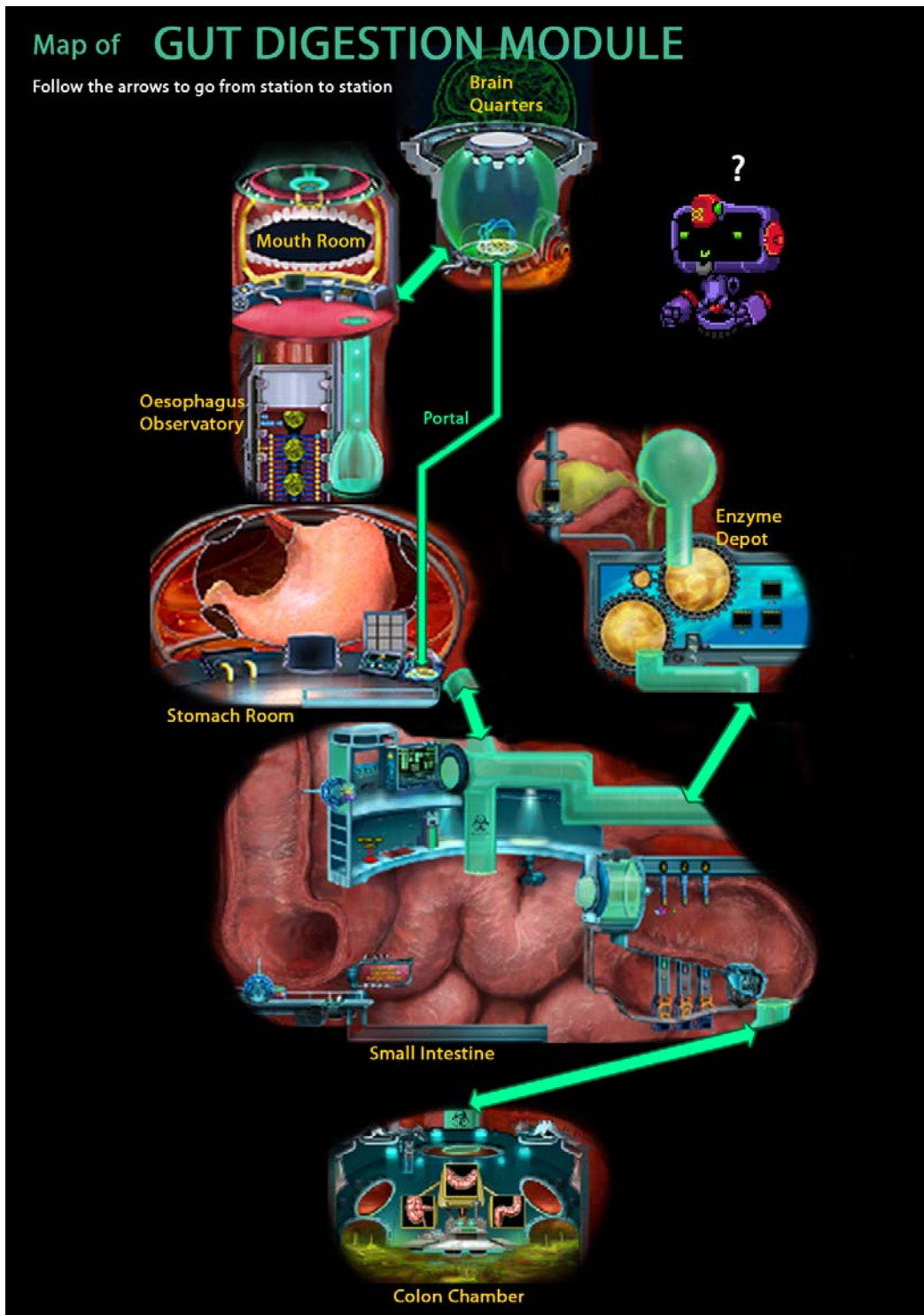


Menu

The menu consists of helpful commands and can be accessed by pressing the <m> key. You can bring up hotseat commands if you should forget them.

Finding your way around

The Control Stations that make up the Gut Replacement Module are laid out to match a real human digestive system. Each station performs one or more essential functions for digestion.



Station 1: Brain Quarters



What can be done here?

The brain quarters are where Automatons help George decide what food he should eat when he gets hungry.

What you can do here

Here, you will plan George's diet with the **food menu**. The food menu shows the food that George can eat. Information on a food is shown when you select it. Once you have chosen to it the food, George will place it into his mouth.

Hotseats available:

- Diet hotseat

Diet Hotseat

Use this hotseat to plan George's Diet and put food into his mouth. You can select a food from the choices available in the food menu.

How to plan George's diet



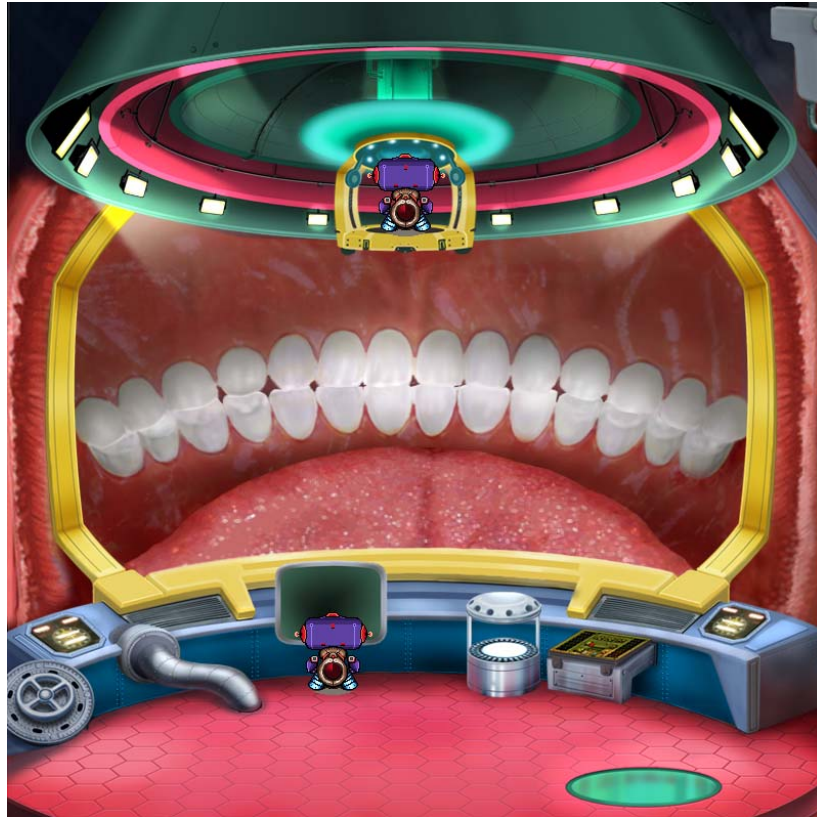
IMPORTANT NOTE:

- You can only eat if no food is present at the mouth. If there is food in the mouth, it must be swallowed first before you can eat another food.
- Plan your diet carefully to fit into the given meal time. Any nutrient that hasn't left the stomach at the end of the scenario will count towards food wasted and impose a penalty on your score.

Cool Facts

The brain is the central command of the human body. It is concerned with intelligence, learning, memory and emotions. All voluntary and involuntary actions are also decided by the brain. **Peristalsis**, an important part of digestion is one of many involuntary actions controlled by a part of the brain known as the **medulla oblongata**. It is located at the back of the brain.

Station 2: Mouth Room



What can be done here?

Many important processes take place at the mouth. These prepare the food for its passage downwards. Food comes in different sizes, hardness or dryness enters the mouth. They are chewed on, softened and shaped into a form suitable for transporting towards the stomach.

What you can do here?

Prepare the food by wetting it with saliva, and chewing it into a ball like shape known as the bolus. If the food contains carbohydrates, an enzyme, **amylase** in the saliva will start digesting it. After swallowing, you can also head to the oesophagus observatory using a green tube at the mouth room.

Hotseats available:

- Saliva Hotseat
- Chewing Hotseat

Cool Facts

The **Mouth** is the beginning of the digestive system. Here, the **teeth** and **tongue** work together, pushing against the walls of the mouth to mix and shape the food in smaller pieces. Our teeth come in four types. They are, from centre to edge of our mouth: incisors (middle 4), canines (1 each side), premolars (2 each side) and molars (2 each side).

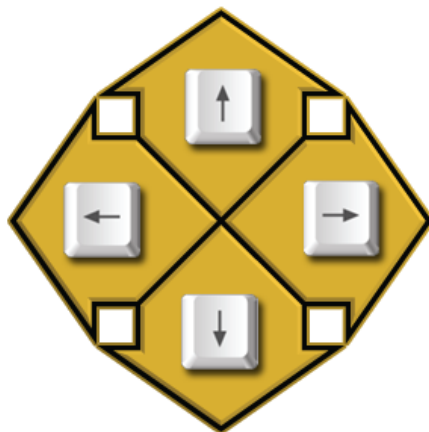
Taste buds found on the tongue enable us to sense and hopefully, enjoy our food. **Saliva** is released by 3 pairs of glands found below our tongue. It is released automatically, but in greater quantity as reaction to our sensing of food when we are hungry or when we place food into our mouth. Note that the enzyme, salivary amylase is found in saliva, which begins to digest carbohydrate in some food. You will notice this when you process food with carbohydrate at the saliva hotseat.

Unlike solid food, fluid based food like water can be swallowed without need for chewing or applying of saliva.

Saliva Hotseat

This hotseat uses an number sequence to show the food's dryness and size. The numbers appear jumbled up on an orange coloured direction indicator like the picture below. The more dry and larger the food, the more saliva you need and the longer the sequence. Finish the sequence to fully wet the food and make digestion easier.

How To Apply Saliva



ORANGE ARROW INDICATOR
& MATCHING DIRECTION KEYS

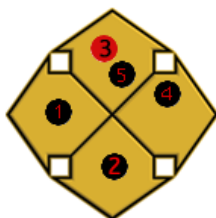
Saliva Application

Press the direction key that matches the arrow where the lowest current number falls on. Finish clearing the number sequence.



Red coloured numbers show carbohydrates that can be digested by enzyme in the saliva, like the '3' in the sequence below

current lowest number is 1 & it falls on left orange arrow



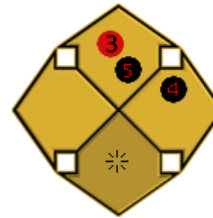
Step 1

current lowest number is 2 & it falls on down orange arrow



Step 2

current lowest number is 3 & it falls on up orange arrow



oh i get it!



Continue clearing until no numbers remain

Chewing Hotseat

Use this control to break and mix food into boli and send them down the oesophagus. You will see a glowing HIGHLIGHT flash along the upper and lower rows of teeth. Stopping this highlight will determine which teeth are used and the strength of your chewing.

How to Chew and Swallow



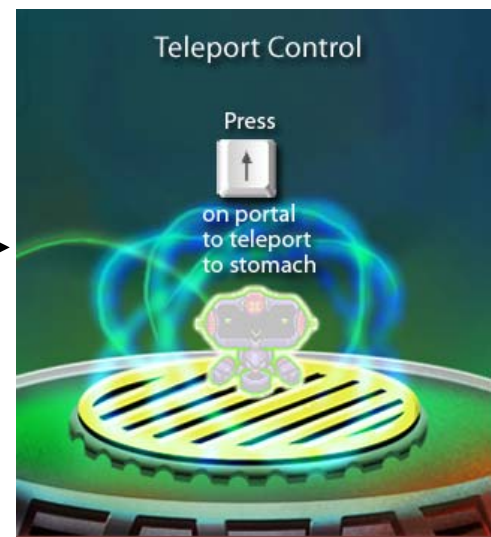
IMPORTANT NOTE:

- To save time, you can chew and apply saliva at the same time. However the chewing strength is maximized (as values shown above) only when saliva is fully applied.
- You cannot swallow food unless some saliva has been applied first.
- Also ensure that there is no more than 2 boli of food waiting before or after the oesophagus.

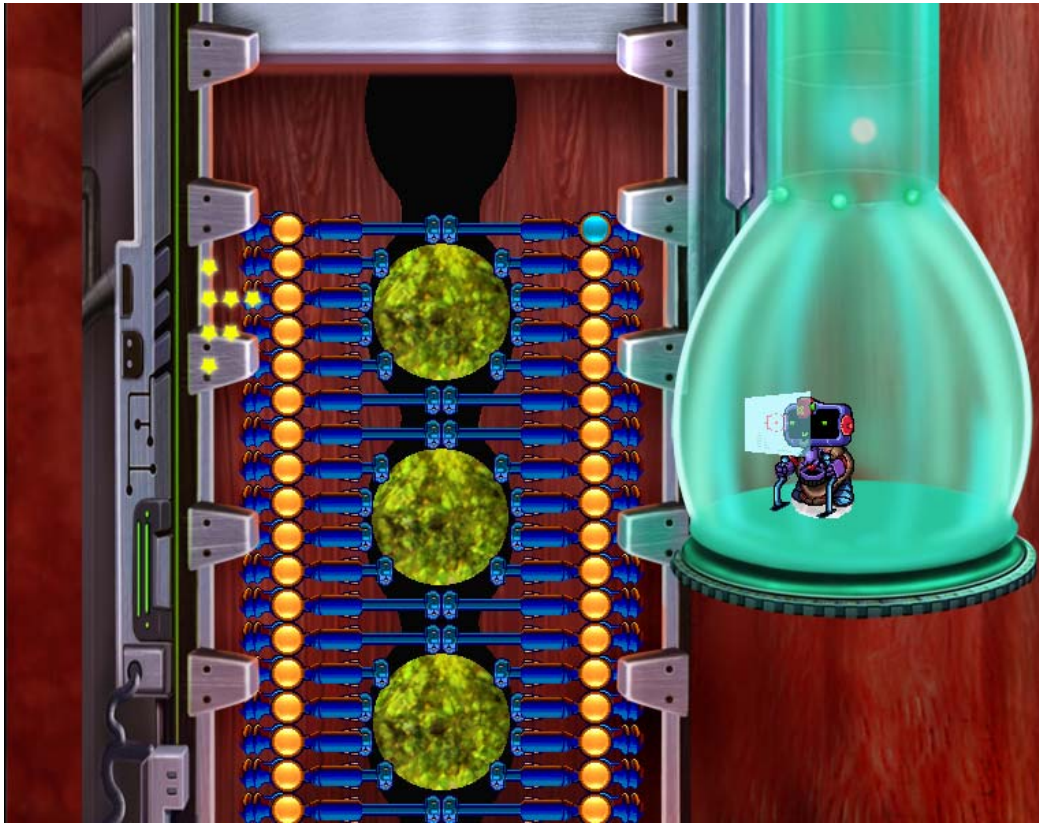
Portal To Stomach

The portal lies below the brain quarters and to the right of the mouth room. It looks like a yellow disc with grills. Use this portal to go to the stomach room.

How to use the Portal to go to the Stomach →



Station 3: Oesophagus Observatory



What can be done here?

The Oesophagus serves as a passage that connects the mouth at the head, to the stomach further down the body. Once food broken down at the mouth, it is moved to the stomach with the help of the oesophagus using a process known as **peristalsis**.

What you can do here

Here, you need to push food towards the stomach by contracting the muscles at a position slightly above the food and relaxing below it.

Hotseats available:

- Peristalsis hotseat

Cool Facts

The **oesophagus** is shaped like a tube, about 30 cm long and lined with muscles on its outer wall. The inner wall of the oesophagus keeps releasing **mucus** to make the wall slippery and prevent food from getting stuck.

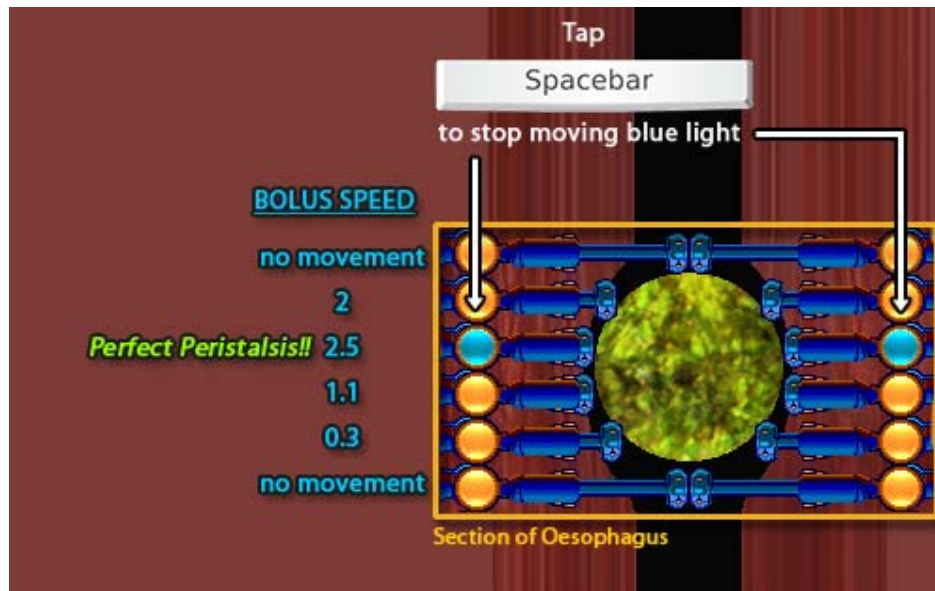
The oesophagus is where peristalsis happens for the first time. **Peristalsis** is the rhythmic tightening and relaxing of muscles to move food along a passage in the digestive system. After swallowing, it takes about less than ten seconds for solid food to be pushed to the stomach. Peristalsis also takes place at the stomach and intestines.

As the mouth is used to take in air as well as food, the oesophagus shares the passage with the **larynx**, which allows air to flow to and from the lungs. To prevent food from entering the larynx, a flap like part known as the **epiglottis** blocks incoming food but allows air to flow into the lungs. Sometimes food gets into the larynx by accident. The body responds automatically by making us cough to expel the food.

Peristalsis Hotseat

Our artificial peristalsis system has 3 sections. Create a rhythm to move the bolus down in the fastest way possible. You need at least 3 contractions and relaxations to send a bolus through the 3 sections.

How to push food down the oesophagus



A section of oesophagus control, showing the six light positions

IMPORTANT NOTE :

- You can have more than one bolus passing through this system at the same time
- You will be awarded a score only when you get 'perfect peristalsis' for this hotseat

Station 4: Slosh n Churn

How Automatons see food from the stomach onwards

Automatons are able to see food as a combination of BLOCKS of nutrients, water and fibre, which we call a **food grid**. This ability helps them process food correctly. Here's what a food grid looks like, and what the symbols mean:

Food Grid of Cheeseburger

																Undigested			
																Partially Digested			
																Fully Digested			
																Indigestible			
C	W	F	C	C	W	C	W	C	W	C	W	C	W	C	W	C			
C	C	W	C	P	F	C	C	C	P	C	C	C	P	C	C				
C	F	C	W	C	C	P	C	P	F	W	P	C	P	C	P				
C	P	W	P	P	F	W	P	P	F	W	P	P	F	W	P				
C	F	C	P	P	F	C	C	C	P	P	F	C	C	C	P				
F	F	F	W	W	W	W	W	W	W	W	W	W	W	W	W				
W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W				

The 'C' stands for carbohydrate.

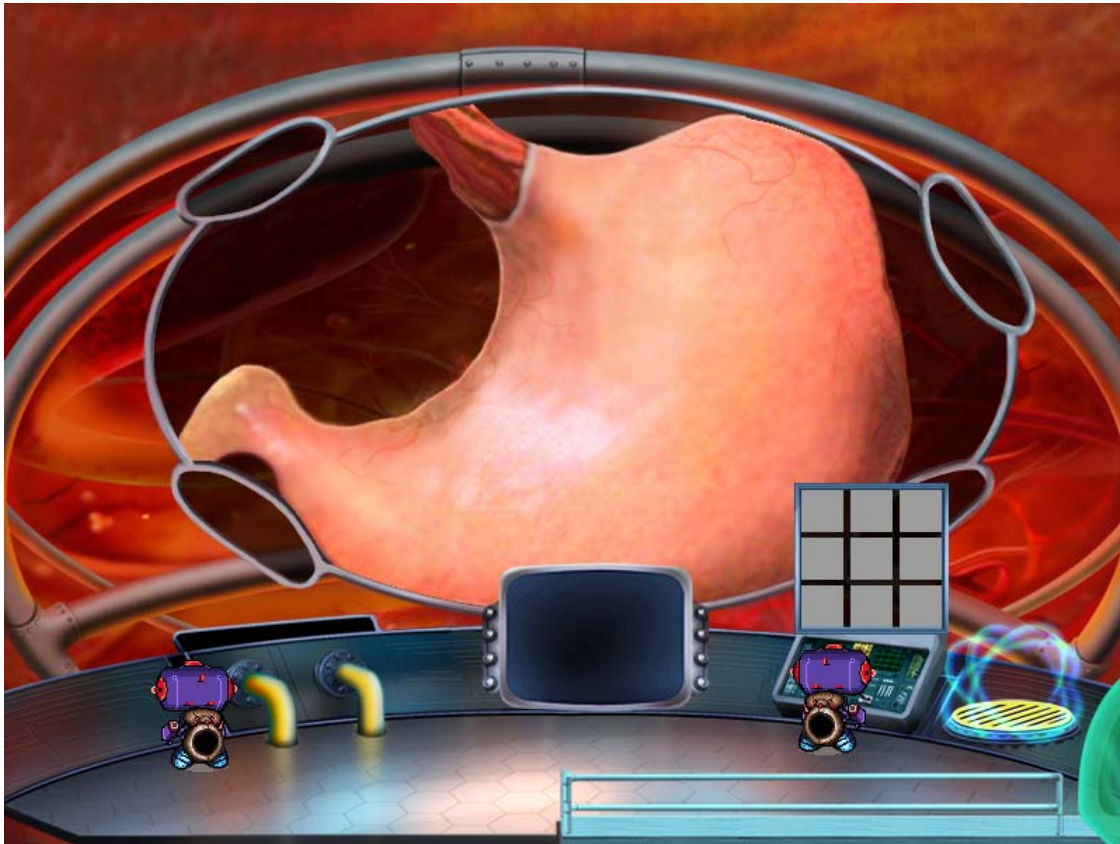
The 'P' stands for protein.

The 'F' stands for fat.

Note that the undigested food have a coloured outer shell, compared to the partially digested food. The block with the water droplet represents water while the brown symbol with curls represents fibre.

IMPORTANT NOTE:

- Food grids may be shown in different sizes, depending on the food chosen.



What can be done here?

Gastric juice containing **Hydrochloric (HCl) acid** is constantly released into the stomach. A protein digesting enzyme called **pepsin** is also released here. This enzyme is released from the stomach wall as you **churn**. It digests protein in food, but only when the acidity in stomach is low enough and if the proteins are exposed on the surface of the food. Churning in the stomach thoroughly breaks up and mixes the food with acid before passing it to the small intestine as liquid like **chyme**.

What you can do here

The acid release system works like a **puzzle**, with a start point and end point, along with various pipes. The pipes break connection after a while to prevent excess acid from being released by accident.

When food enters the stomach, lower the pH level to 2 to activate the enzyme. Do this by solving the puzzle twice.

Use the churning control to thoroughly break food into a form suitable for digestion. When this happens, the stomach will automatically release the food into the small intestine.

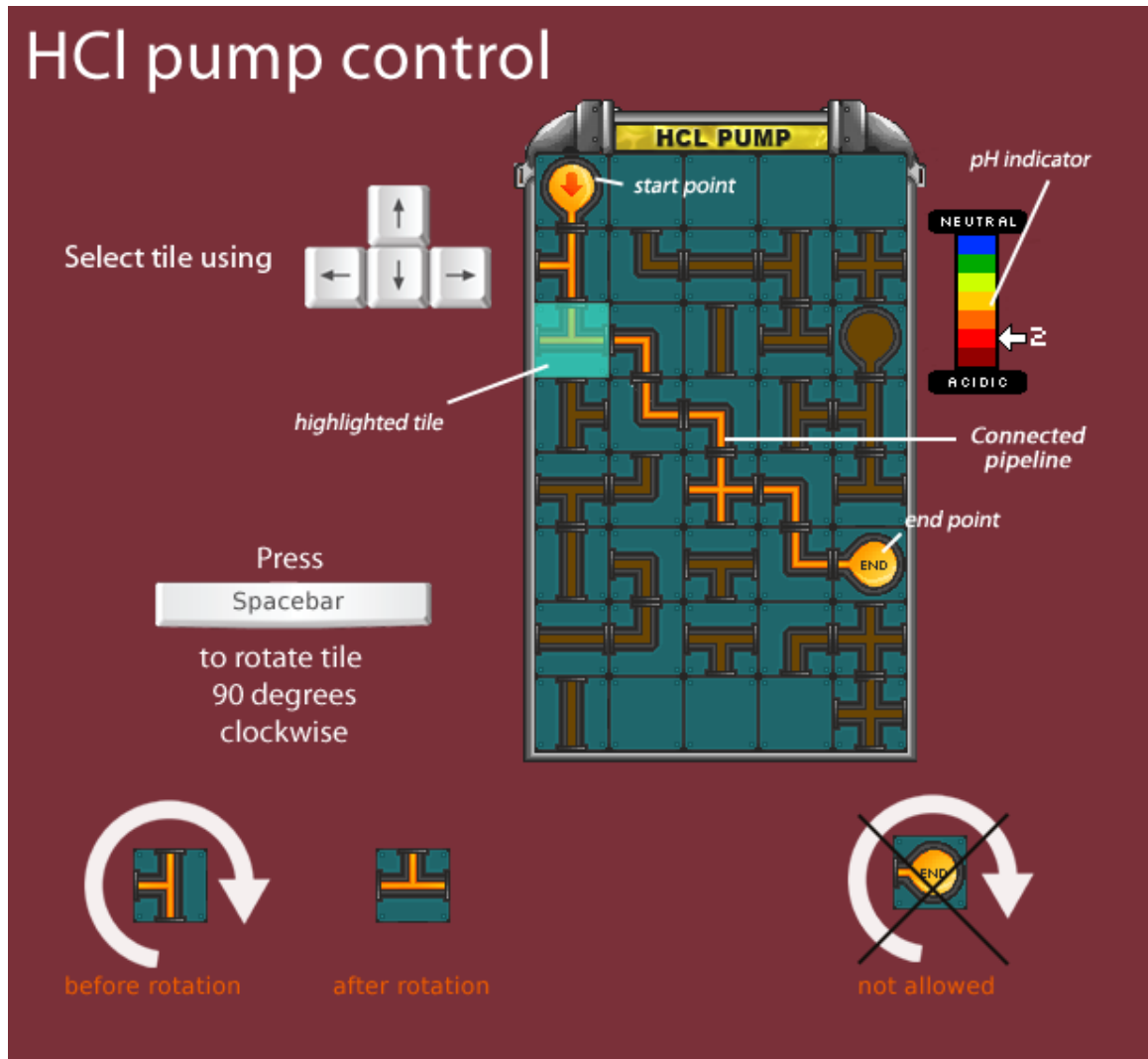
Hotseats available:

- pH hotseat
- Churning and Pepsin hotseat

pH Hotseat

Form a continuous pipeline by connecting the start point to the end point. Do this twice.

How to release acid in the stomach



Lower the pH level down to 2. This activates the enzyme, pepsin which will digest exposed proteins if any.

IMPORTANT NOTE :

- You cannot operate the controls again until 3 minutes has passed, when the pH value in the stomach returns to 4.

Churning and Pepsin Hotseat

Catch the rhythm of the natural peristalsis by getting the same 3 symbols in a row. The better the alignment, the stronger the effect of peristalsis and the faster the food grid gets broken up. When the food grid is reduced to a certain point, it is automatically released from the stomach.

How to churn food and release pepsin in the stomach



IMPORTANT NOTE :

- At the churning hotseat, if you did not reduce the hardness of the food fully at the mouth, the stomach will have to do extra work to reduce the hardness first.

Cool Facts

The stomach is a thick muscular bag. The walls of the stomach are coated with **mucus** and have many gastric glands. The gastric glands constantly release **gastric juice**, which contain hydrochloric acid. When food enters the stomach, the amount of acid is increased. The level of acidity is measured with a unit known as pH. The lower the pH, the more acidic the content. The higher the pH, the more alkaline the content. Because of the acidic nature of gastric juice, the stomach protects itself by constantly secreting mucus at a rate greater than that of erosion caused by acid. However, in certain cases, excess hydrochloric acid may 'burn' through the mucus and erode parts of the stomach wall, causing painful gastric ulcers.

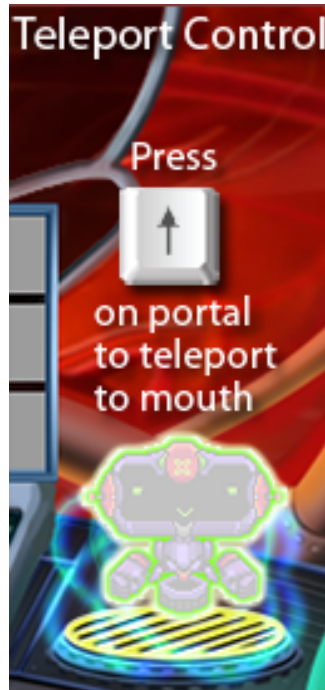
Gastric juice also contains the protein digesting enzymes pepsin and rennin. Both are activated by hydrochloric acid. Before activation, **pepsin** is in a form known as pepsinogen. **Rennin** is used to curdle milk proteins by converting them from soluble caseinogen into insoluble casein. By making the protein insoluble, the stomach prevents them from escaping digestion by pepsin.

The entry and exit to the stomach is controlled by valves. The valves open to let food into the stomach or out of it. They close up when the stomach churns food, which will last for about 3 to 4 hours. Food then exits the stomach from a valve known as the **pyloric sphincter** and heads into the small intestine. However, sickness or bad food may sometimes cause the stomach to empty the content through the mouth, through contractions of the abdomen.

Portal to Mouth Area:

A small floating yellow disc-like portal can be found in this room. Use it to go to the mouth and brain room by:

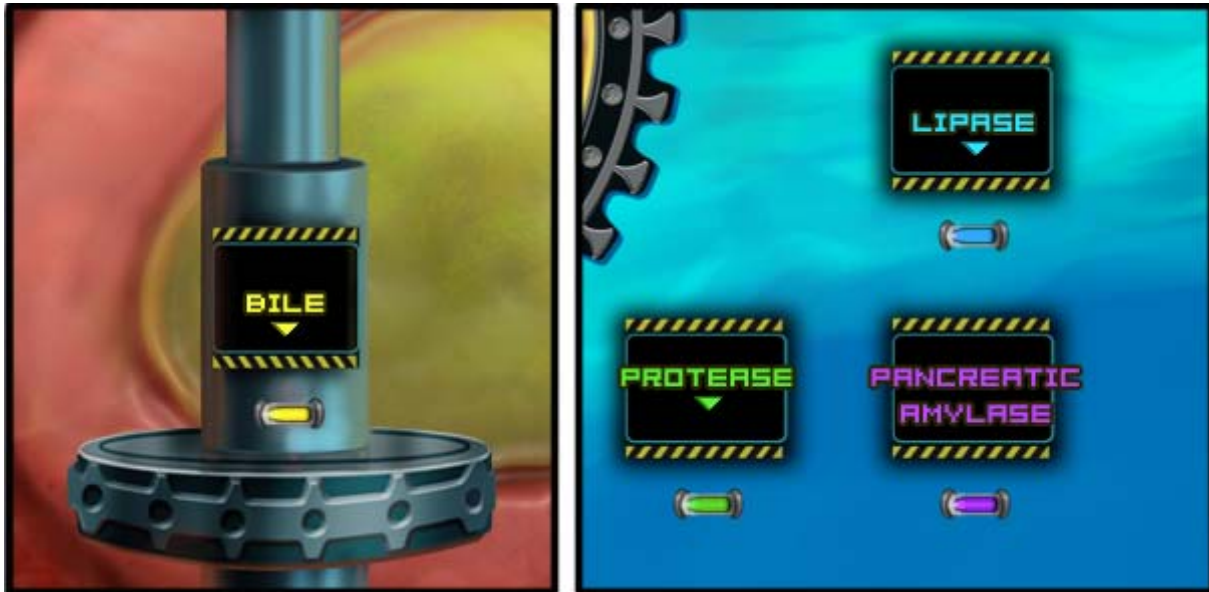
How to use the portal to go to the mouth



IMPORTANT NOTE:

- Unlike the normal stomach, George's new system does not allow more than one bolus inside the stomach at any one time.

Station 5: Enzyme Depot

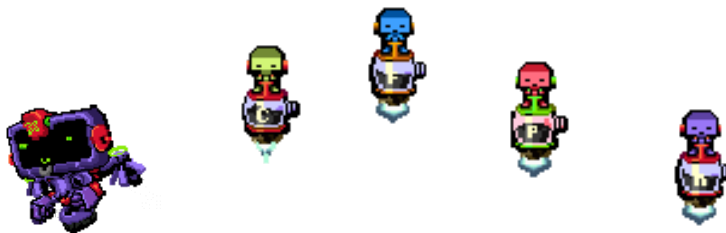


What can be done here?

The Enzyme Depot stores enzymes produced by George's Pancreas. It also stores Bile, produced at the Liver and collected at the Gall Bladder. The Pancreatic enzymes are needed for the process of digesting food into simpler substances that can be absorbed for use by the body. The bile is used to **emulsify** fat molecules to make it easier for digestion by the fat enzyme.

What you can do here

The enzymes found here are for use with **Turret 1**. You can collect them by getting the help of the enzyme helpers. Lead them to the enzyme loader, then activate the loading button.



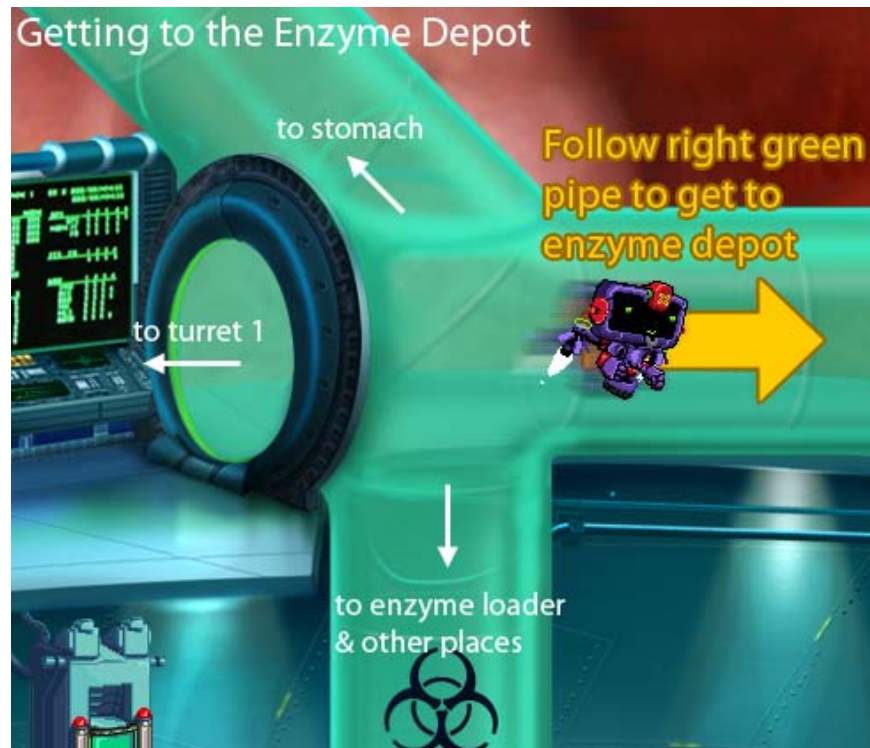
Enzyme Helpers for (l-r) carbohydrate, fat, protein, bile following an Automaton

Hotseats available

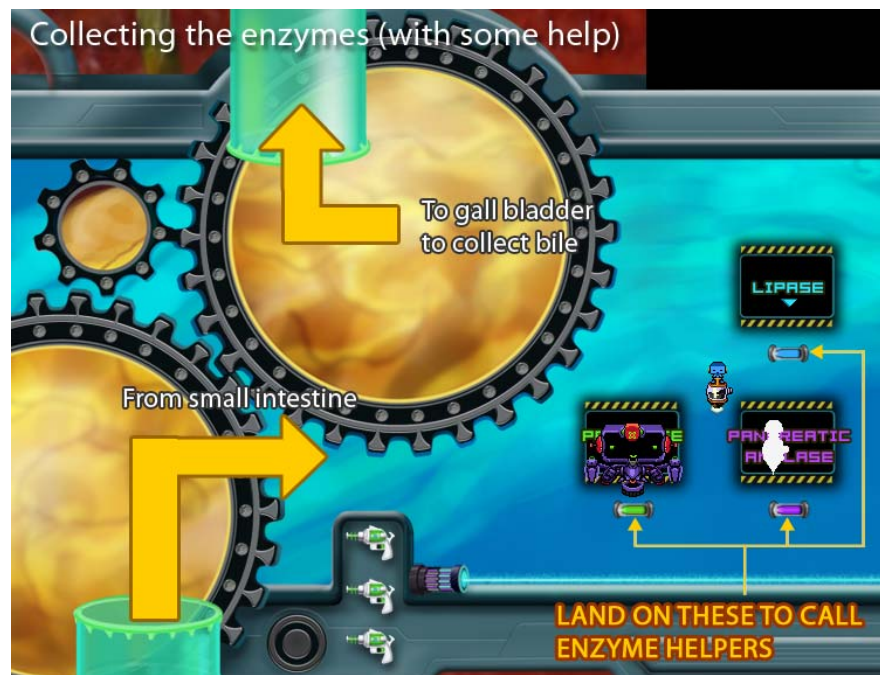
- None

How to collect enzymes for turret 1

Step 1:



Step 2:



How to collect enzymes for turret 1

Step 3:



IMPORTANT NOTE:

- Do not fly out of the small intestine or enzyme depot with the helpers in tow or they will get lost and fly home. You will then have to summon them again.
- You need not stand on the load enzyme button to finish the loading.
- Each enzyme helper carries 30 enzymes per trip.

Cool Facts

The liver is the largest gland in the human body. It is dark red in colour and holds a lot of blood. It secretes a liquid called bile and store it in a small bag known as the gall bladder.

The Pancreas is a fleshy looking gland that produces many important enzymes necessary for digestion. The Pancreas and Liver have other very important roles to play in the body.

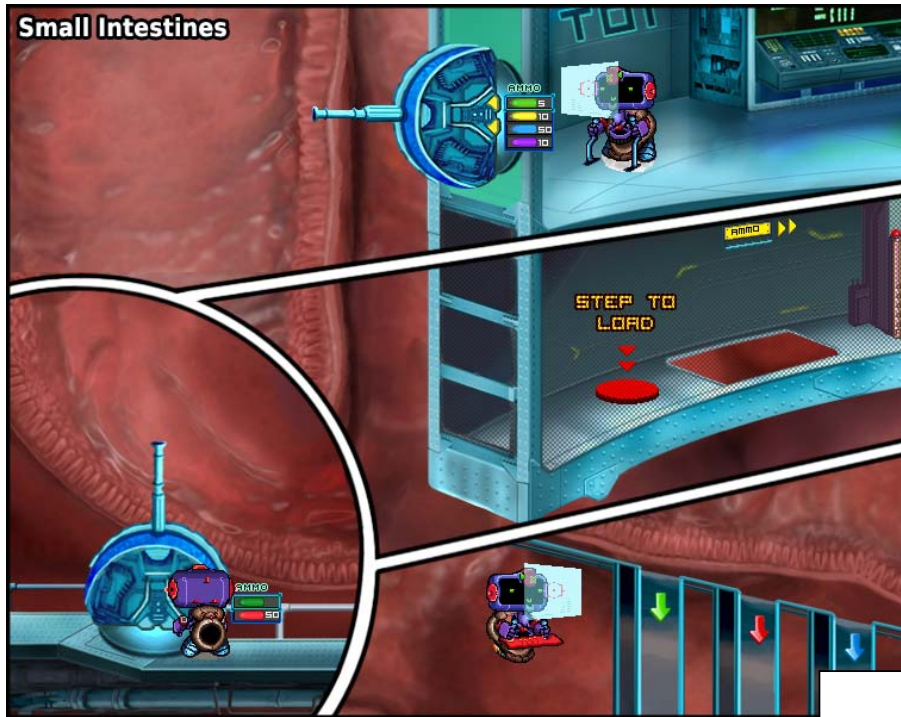
For instance, the pancreas creates an important hormone, insulin that works with the liver to keep the amount of sugar in our blood at a healthy level.

One of the functions of the liver is to produce a yellow greenish substance known as bile. The liver sends bile to the gall bladder, where it is stored until release for the emulsification of fat.

Emulsification is a chemical process which breaks up fat in food into globules making fat digestion easier for the fat enzymes.

Do you know that dried bile pigments are responsible for gives faeces its foul looking colour?

Station 6: Small Intestine Room



What can be done here?

After food exits the stomach, it will arrive at the small intestine. Many enzymes are released into the intestine. These are released along the intestinal wall. Before exiting the small intestine, the enzymes would have acted on protein, fat and carbohydrate, reducing them into forms that can be readily absorbed at the small intestine.

What you can do here

Use the two turrets found along the small intestine to release enzymes to fully digest the three main organic nutrients. Food that's been fully broken down into nutrients can now be sent to parts of the body that needs them. Use the Absorption Puncher that has been put further down George's small intestine to do this.

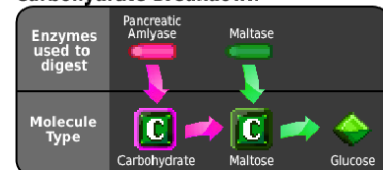
You can also load up enzymes at the enzyme loader. It is found below turret 1.

Hotseats available

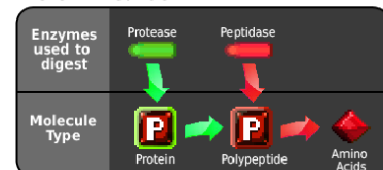
- Turret 1, pancreatic enzyme
- Turret 2, Intestinal enzyme
- Absorption puncher

How food is digested for absorption

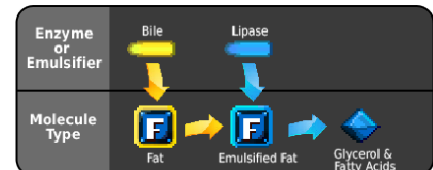
Carbohydrate Breakdown



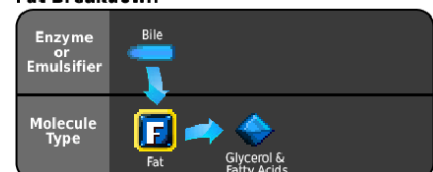
Protein Breakdown



Fat Emulsification & Breakdown



Fat Breakdown

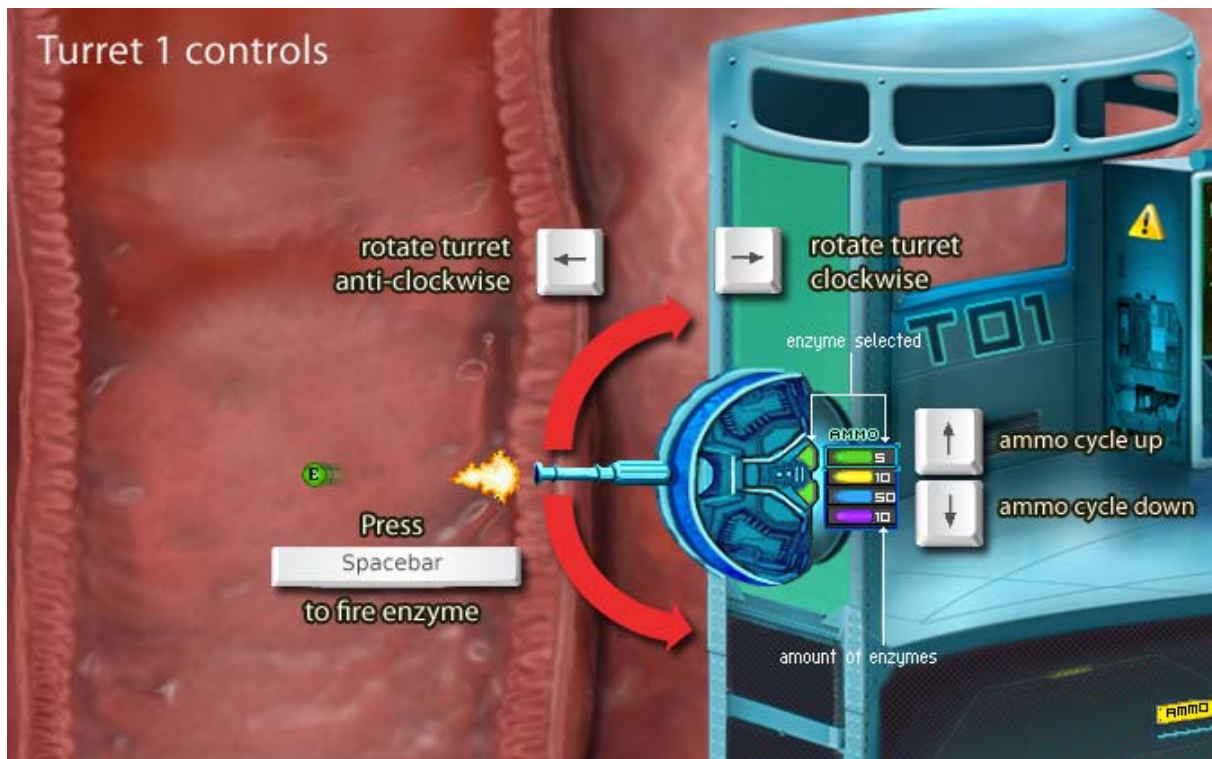


Enzyme turret 1 Hotseat, Pancreatic Enzymes

Use turret 1 to:

- Digest complex carbohydrate into simpler form **for turret 2**
- Digest complex protein into simpler form **for turret 2**
- Emulsify fat for easier digestion using enzyme bile
- Digest emulsified fat into simplest form
- Digest fat without emulsification into simplest form for absorption.

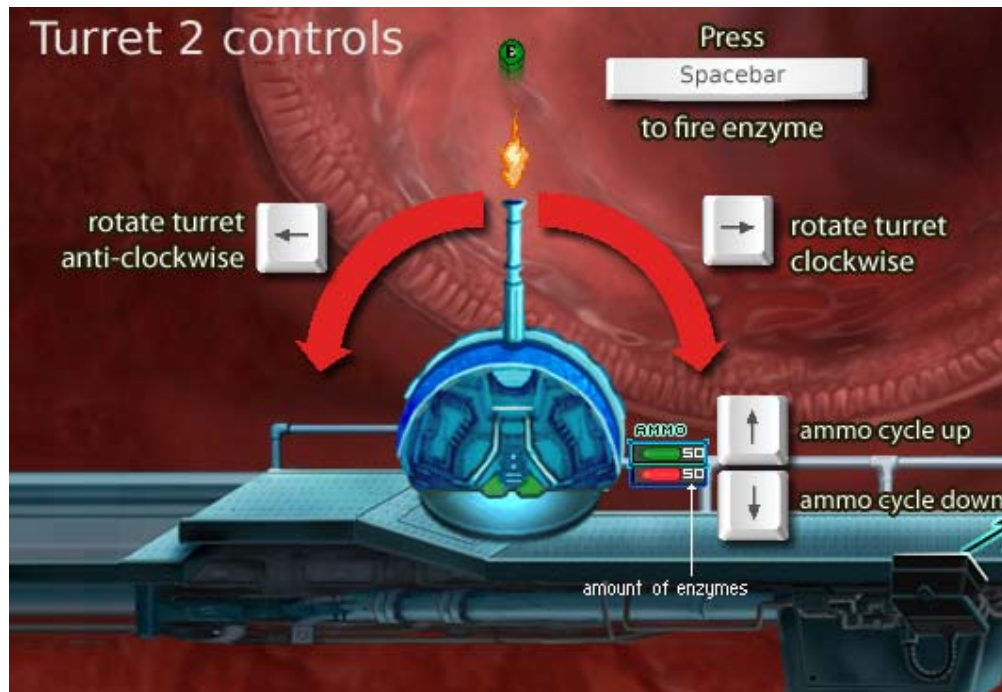
How to operate Turret 1



Enzyme Turret 2 Hotseat, Intestinal Enzymes

Use Turret 2 to **further digest** proteins and carbohydrates into their simplest form.

How to operate Turret 2



IMPORTANT NOTE:

- You do not need to collect enzymes for turret 2 as these are produced and taken from the intestinal mucosa.

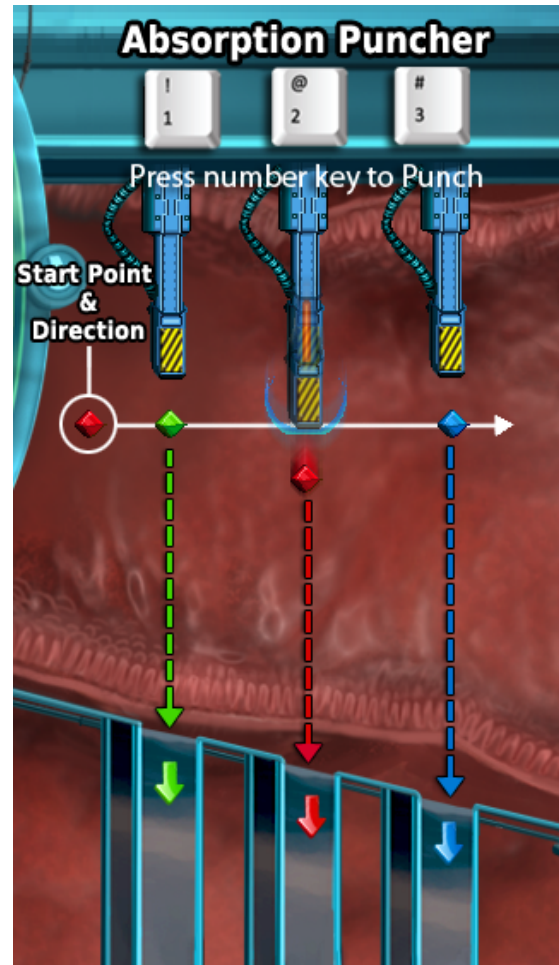
Absorption Hotseat

Send the nutrient down the proper pipe as it passes by:

How to absorb digested food

IMPORTANT NOTE:

- Only fully digested food is small enough to be absorbed here. Sending digested food down the wrong pipe will give you lesser points.
- Do not send undigested food, partially digested food or fibre down the pipes or you will be penalised!
- Most of the water is automatically absorbed at the Colon. You can absorb water at this hotseat but you will not be rewarded for your efforts. Furthermore, the colon performs this much better.



Cool Facts

The small intestine consists of a U-shaped duodenum, followed by the jejunum and ends with the highly coiled ileum. All in all, our small intestine is about six metres long. A great deal of digestion takes place here. The turret 1 in the game is located where the pancreatic duct joins the duodenum. It is where pancreatic juice is released. Turret 2 comes after turret 1 to finish of the digestion process. In healthy bodies, the role of both turret 2 and the absorption control is performed by the wall of the small intestine. The surface of the small intestine is covered in folds known as villi, which in turn, are covered in smaller folds known as microvilli. These folds help to increase the area of contact between the food and the surface of the small intestine. Food is digested to its simplest form by enzymes on the inner surface of the small intestine, before being absorbed through the microvilli into our bloodstream. This special characteristic, along with the length of the small intestine allows us to make the most of the food we eat by increasing digestion and absorption. After the nutrients are absorbed, a process known as assimilation takes place. Your body will decide what to do with these nutrients. Some will be changed into materials that will help you grow or replace worn out parts in your cells. Others may be converted into energy that you need to carry out your body's activities, or stored for use at a later time. Some nutrients also help your body stay healthy.

Station 7: Colon Chamber



What is done here?

At the colon, water, minerals and vitamins are absorbed, leaving undigested material and waste products to be removed from the body. There are no controls present in this station.

Cool Facts

The large intestine carries on from where the small intestine ends, beginning on the right side within your abdomen. What's left of the food you've eaten will pass upwards, then sideways before moving downwards on the left side and exiting through the anus as faeces. The large intestine, as its name suggests is larger than the small intestine but shorter about 1.5 metres. It is important for the absorption of vitamins and mineral salts, and most of the water that you drink. It also holds undigested material for a period of time, until you feel the urge to pass them out of your body, along with some bacteria.

Crisis? What Crisis?

Crises may come up if there is poor teamwork. These happen when food is not being taken care of in a room. You will see an area in George's minimap flash red during a crisis.

Causing food queues before the oesophagus or before the stomach to exceed their limit will create a more serious crisis. When this happens, the control stations affected will shutdown until you have fixed the problem. During shutdown, there will also be a score penalty.

NUTRITIONAL INFORMATION

The organic nutrients that automatons see are in the form of blocks. **One block holds 3g of a nutrient.**

The normal human require about 2000-2200 CALORIES (or kcal) of food intake a day.

George's ideal intake is ~2100 CALORIES a day.
 His daily recommended nutrition in blocks is: 96
 carbohydrates: 24 fats: 24 proteins

Try to keep within reasonable limits of the recommended intake to keep George in good health ie not less than 1622 CALORIES DAILY & not more than 2359 CALORIES DAILY.
 Data of George's food diet is in the following pages for your reference.

1g of protein will yield	4 CALORIES(kcal) of energy
1g of carbohydrate will yield	4 CALORIES(kcal) of energy
1g of fat will yield	9 CALORIES (kcal) of energy

Food Data

Baked Beans  18% 80% 2% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>1</td><td>3g</td></tr> <tr><td>Carbohydrate</td><td>3</td><td>9g</td></tr> <tr><td>Fat</td><td>0</td><td>0g</td></tr> <tr><td>Fibre</td><td>1</td><td>3g</td></tr> <tr><td>Water</td><td>11</td><td>33g</td></tr> <tr><td>Total:</td><td>16</td><td>48g</td></tr> </table> Hardness Value: 450 A good source of dietary fiber, for healthy bowel function.		Blocks	Qty	Protein	1	3g	Carbohydrate	3	9g	Fat	0	0g	Fibre	1	3g	Water	11	33g	Total:	16	48g	Curry Chicken  69% 1% 30% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>17</td><td>51g</td></tr> <tr><td>Carbohydrate</td><td>0</td><td>0g</td></tr> <tr><td>Fat</td><td>4</td><td>12g</td></tr> <tr><td>Fibre</td><td>0</td><td>0g</td></tr> <tr><td>Water</td><td>43</td><td>129g</td></tr> <tr><td>Total:</td><td>64</td><td>192g</td></tr> </table> Hardness Value: 950 Chicken seasoned with thick curry gravy. It makes a tasty meal.		Blocks	Qty	Protein	17	51g	Carbohydrate	0	0g	Fat	4	12g	Fibre	0	0g	Water	43	129g	Total:	64	192g	Mixed Vegetables Dish  12% 86% 2% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>2</td><td>6g</td></tr> <tr><td>Carbohydrate</td><td>8</td><td>24g</td></tr> <tr><td>Fat</td><td>0</td><td>0g</td></tr> <tr><td>Fibre</td><td>3</td><td>9g</td></tr> <tr><td>Water</td><td>43</td><td>129g</td></tr> <tr><td>Total:</td><td>51</td><td>153g</td></tr> </table> Hardness Value: 900 For the health-conscious, this dish provides the necessary nutrition your body needs.		Blocks	Qty	Protein	2	6g	Carbohydrate	8	24g	Fat	0	0g	Fibre	3	9g	Water	43	129g	Total:	51	153g	Steamed Snapper Fish  88% 0% 12% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>17</td><td>51g</td></tr> <tr><td>Carbohydrate</td><td>0</td><td>0g</td></tr> <tr><td>Fat</td><td>1</td><td>3g</td></tr> <tr><td>Fibre</td><td>0</td><td>0g</td></tr> <tr><td>Water</td><td>46</td><td>138g</td></tr> <tr><td>Total:</td><td>64</td><td>192g</td></tr> </table> Hardness Value: 600 An excellent source of omega 3 fatty acids. It improves the health of our brain and eyes.		Blocks	Qty	Protein	17	51g	Carbohydrate	0	0g	Fat	1	3g	Fibre	0	0g	Water	46	138g	Total:	64	192g
	Blocks	Qty																																																																																					
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Carbohydrate	3	9g																																																																																					
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Fruity Muesli Bar  11% 78% 13% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>2</td><td>6g</td></tr> <tr><td>Carbohydrate</td><td>12</td><td>36g</td></tr> <tr><td>Fat</td><td>1</td><td>3g</td></tr> <tr><td>Fibre</td><td>1</td><td>3g</td></tr> <tr><td>Water</td><td>0</td><td>0g</td></tr> <tr><td>Total:</td><td>16</td><td>48g</td></tr> </table> Hardness Value: 850 Yummy and nutritious light snack made up of whole grains and various fruits.		Blocks	Qty	Protein	2	6g	Carbohydrate	12	36g	Fat	1	3g	Fibre	1	3g	Water	0	0g	Total:	16	48g	Tropical Jawbreaker  0% 100% 0% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>0</td><td>0g</td></tr> <tr><td>Carbohydrate</td><td>8</td><td>24g</td></tr> <tr><td>Fat</td><td>0</td><td>0g</td></tr> <tr><td>Fibre</td><td>0</td><td>0g</td></tr> <tr><td>Water</td><td>0</td><td>0g</td></tr> <tr><td>Total:</td><td>8</td><td>24g</td></tr> </table> Hardness Value: 1000 Delightful sweets to satisfy your sweet tooth craving! Keeps your mouth busy all day.		Blocks	Qty	Protein	0	0g	Carbohydrate	8	24g	Fat	0	0g	Fibre	0	0g	Water	0	0g	Total:	8	24g	Fried Egg  35% 9% 62% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>2</td><td>6g</td></tr> <tr><td>Carbohydrate</td><td>0</td><td>0g</td></tr> <tr><td>Fat</td><td>2</td><td>6g</td></tr> <tr><td>Fibre</td><td>0</td><td>0g</td></tr> <tr><td>Water</td><td>12</td><td>36g</td></tr> <tr><td>Total:</td><td>16</td><td>48g</td></tr> </table> Hardness Value: 400 Provide nutrients that help regulate cholesterol level and develop muscles.		Blocks	Qty	Protein	2	6g	Carbohydrate	0	0g	Fat	2	6g	Fibre	0	0g	Water	12	36g	Total:	16	48g	Chocolate Bar  4% 48% 56% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>1</td><td>3g</td></tr> <tr><td>Carbohydrate</td><td>9</td><td>27g</td></tr> <tr><td>Fat</td><td>4</td><td>12g</td></tr> <tr><td>Fibre</td><td>1</td><td>3g</td></tr> <tr><td>Water</td><td>1</td><td>3g</td></tr> <tr><td>Total:</td><td>16</td><td>48g</td></tr> </table> Hardness Value: 800 Has antioxidants (flavonoids) which help protect against free radicals that cause cell and tissue damage.		Blocks	Qty	Protein	1	3g	Carbohydrate	9	27g	Fat	4	12g	Fibre	1	3g	Water	1	3g	Total:	16	48g
	Blocks	Qty																																																																																					
Protein	2	6g																																																																																					
Carbohydrate	12	36g																																																																																					
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Carbohydrate	8	24g																																																																																					
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Water	1	3g																																																																																					
Total:	16	48g																																																																																					
Banana  4% 93% 3% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>0</td><td>0g</td></tr> <tr><td>Carbohydrate</td><td>8</td><td>24g</td></tr> <tr><td>Fat</td><td>0</td><td>0g</td></tr> <tr><td>Fibre</td><td>1</td><td>3g</td></tr> <tr><td>Water</td><td>23</td><td>69g</td></tr> <tr><td>Total:</td><td>16</td><td>48g</td></tr> </table> Hardness Value: 400 An excellent source of potassium and fibre which helps in our digestion.		Blocks	Qty	Protein	0	0g	Carbohydrate	8	24g	Fat	0	0g	Fibre	1	3g	Water	23	69g	Total:	16	48g	Red Apple  0% 100% 0% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>0</td><td>0g</td></tr> <tr><td>Carbohydrate</td><td>1</td><td>3g</td></tr> <tr><td>Fat</td><td>0</td><td>0g</td></tr> <tr><td>Fibre</td><td>1</td><td>3g</td></tr> <tr><td>Water</td><td>6</td><td>18g</td></tr> <tr><td>Total:</td><td>8</td><td>24g</td></tr> </table> Hardness Value: 800 An excellent source of antioxidants and able to relieve constipation.		Blocks	Qty	Protein	0	0g	Carbohydrate	1	3g	Fat	0	0g	Fibre	1	3g	Water	6	18g	Total:	8	24g	Cookie Chips  5% 65% 30% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>1</td><td>3g</td></tr> <tr><td>Carbohydrate</td><td>5</td><td>15g</td></tr> <tr><td>Fat</td><td>2</td><td>6g</td></tr> <tr><td>Fibre</td><td>0</td><td>0g</td></tr> <tr><td>Water</td><td>0</td><td>0g</td></tr> <tr><td>Total:</td><td>8</td><td>24g</td></tr> </table> Hardness Value: 600 Huge crunchy cookies that makes a worthy light-hearted snack.		Blocks	Qty	Protein	1	3g	Carbohydrate	5	15g	Fat	2	6g	Fibre	0	0g	Water	0	0g	Total:	8	24g	French Fries  4% 48% 48% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>1</td><td>3g</td></tr> <tr><td>Carbohydrate</td><td>12</td><td>36g</td></tr> <tr><td>Fat</td><td>6</td><td>18g</td></tr> <tr><td>Fibre</td><td>1</td><td>3g</td></tr> <tr><td>Water</td><td>12</td><td>36g</td></tr> <tr><td>Total:</td><td>32</td><td>96g</td></tr> </table> Hardness Value: 300 Looking for a quick snack? This crispy golden fries is just what you need!		Blocks	Qty	Protein	1	3g	Carbohydrate	12	36g	Fat	6	18g	Fibre	1	3g	Water	12	36g	Total:	32	96g
	Blocks	Qty																																																																																					
Protein	0	0g																																																																																					
Carbohydrate	8	24g																																																																																					
Fat	0	0g																																																																																					
Fibre	1	3g																																																																																					
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Protein	0	0g																																																																																					
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Protein	1	3g																																																																																					
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Fibre	1	3g																																																																																					
Water	12	36g																																																																																					
Total:	32	96g																																																																																					
Bowl of Rice  7% 92% 7% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>4</td><td>12g</td></tr> <tr><td>Carbohydrate</td><td>51</td><td>153g</td></tr> <tr><td>Fat</td><td>0</td><td>0g</td></tr> <tr><td>Fibre</td><td>1</td><td>3g</td></tr> <tr><td>Water</td><td>8</td><td>24g</td></tr> <tr><td>Total:</td><td>64</td><td>192g</td></tr> </table> Hardness Value: 600 Our main staple of food! Has great source of complex carbohydrates, to give our body energy.		Blocks	Qty	Protein	4	12g	Carbohydrate	51	153g	Fat	0	0g	Fibre	1	3g	Water	8	24g	Total:	64	192g	Dried Noodles  6% 45% 49% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>2</td><td>6g</td></tr> <tr><td>Carbohydrate</td><td>18</td><td>54g</td></tr> <tr><td>Fat</td><td>9</td><td>27g</td></tr> <tr><td>Fibre</td><td>1</td><td>3g</td></tr> <tr><td>Water</td><td>2</td><td>6g</td></tr> <tr><td>Total:</td><td>32</td><td>96g</td></tr> </table> Hardness Value: 600 It is filling and easy to digest. However, they are low in other nutrients.		Blocks	Qty	Protein	2	6g	Carbohydrate	18	54g	Fat	9	27g	Fibre	1	3g	Water	2	6g	Total:	32	96g	Strawberry Yogurt  16% 77% 7% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>1</td><td>3g</td></tr> <tr><td>Carbohydrate</td><td>6</td><td>18g</td></tr> <tr><td>Fat</td><td>0</td><td>0g</td></tr> <tr><td>Fibre</td><td>0</td><td>0g</td></tr> <tr><td>Water</td><td>25</td><td>75g</td></tr> <tr><td>Total:</td><td>32</td><td>96g</td></tr> </table> Hardness Value: 100 Makes a healthy and tasty dessert!		Blocks	Qty	Protein	1	3g	Carbohydrate	6	18g	Fat	0	0g	Fibre	0	0g	Water	25	75g	Total:	32	96g	Cheese Burger  23% 42% 35% Protein Carbo Fats Nutrition Facts <table> <tr><th></th><th>Blocks</th><th>Qty</th></tr> <tr><td>Protein</td><td>8</td><td>24g</td></tr> <tr><td>Carbohydrate</td><td>19</td><td>57g</td></tr> <tr><td>Fat</td><td>8</td><td>24g</td></tr> <tr><td>Fibre</td><td>1</td><td>3g</td></tr> <tr><td>Water</td><td>28</td><td>84g</td></tr> <tr><td>Total:</td><td>64</td><td>192g</td></tr> </table> Hardness Value: 1000 Tasty but has a high amount of carbohydrate and calories enough to fill an empty stomach.		Blocks	Qty	Protein	8	24g	Carbohydrate	19	57g	Fat	8	24g	Fibre	1	3g	Water	28	84g	Total:	64	192g
	Blocks	Qty																																																																																					
Protein	4	12g																																																																																					
Carbohydrate	51	153g																																																																																					
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ABOUT THE DEVELOPER

